

# SMART INDIA HACKATHON 2026



## PashuPoshan AI (पशु-पोषण AI)

Smart AI-Enabled Rapid Cattle Feed & Silage Quality Testing System

● **Theme:** Agriculture, FoodTech & Rural Development

● **Problem Statement:** [PS ID: 26111](#)

Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers

● **Ministry / Department:** Ministry of Fisheries, Animal Husbandry & Dairying (DAHD)

● **PS Category:** [Software Edition](#) (Edge AI Vision + Offline Mobile PWA)

● **Team Name:** PashuPoshan AI

● **Team Leader:** [Pulkit Kumar](#)

● **Team Members & Domain Roles:**

1. **Pulkit Kumar**

*Team Lead & Edge ML*

2. **Dairy Science Lead**

*Ruminant Nutrition & BIS*

3. **Computer Vision Lead**

*CIEDE2000 & MobileNet*

4. **Cloud & PWA Lead**

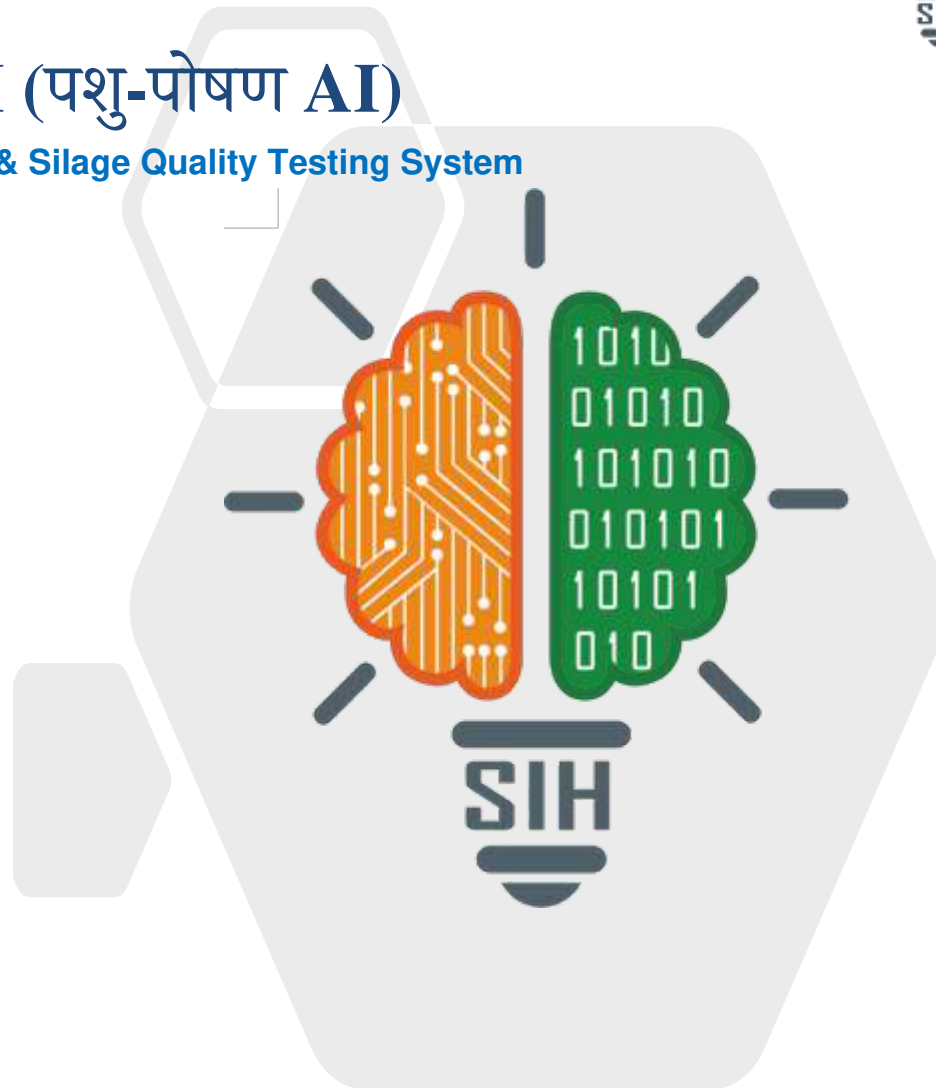
*Offline Architecture & IDB*

5. **Vernacular UX Lead**

*Voice Synthesis & A11y*

6. **Field Validation Lead**

*Farmer Testing & QA*



## ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

### Pointer 1

#### Detailed Explanation of Proposed Solution

- **Zero-Hardware Rapid Smartphone Testing:** Instant on-farm feed and silage diagnostics using phone cameras without lab equipment.
- **Bimodal Testing Protocol:**
  - *Silage Quality:* Flieg Score computation via physical texture + CIEDE2000 pH strip colorimetry.
  - *Dry Feed / Pellets:* Verifies BIS IS:2052 compliance (Moisture, Crude Protein, Fiber).
- **Dynamic Spoilage Detection:** Statistical relative color clustering flags fungal/mold anomalies (cottony white, olive penicillium) before feeding cattle.

**Output:** Interactive Tier A/B/C Scorecard, Printable PDF Certificate & Vernacular Voice Guidance.

### Pointer 2

#### How It Addresses the Problem

- **Overcomes Lab Barrier:** Replaces slow 7–14 day wet-chemistry labs (costing >₹1,500/sample) with a **30-second on-farm test at <₹0.50/scan**.
- **Solves the Rural Connectivity Divide:** *Offline-First Architecture* via Service Worker & IndexedDB guarantees full triage even with 0 kbps internet in remote sheds.
- **Prevents Severe Livestock Losses:** Early detection halts sub-acute ruminal acidosis (SARA) and stops carcinogenic *Aflatoxin M1* from entering the retail milk supply.
- **Overcomes Illiteracy:** Audio advisories in 5 Indic languages (Hindi, Marathi, Gujarati, etc.).

**Impact:** Protects 80M+ smallholder farmers from devastating animal mortality & milk rejection.

### Pointer 3

#### Innovation and Uniqueness

- **Edge-Guarded Dual Triage:** Client-side MobileNetV1 (~1.8MB weights) filters non-feed images in 30ms before invoking cloud AI, saving bandwidth & API costs.
- **CIEDE2000 Colorimetry:** Reference-calibrated delta-E metric neutralizes erratic ambient barn lighting and cheap phone sensor biases.
- **Safety-Capped Conservative Triage:** Unconfirmed offline scans are strictly capped at Tier B (Caution), avoiding costly false-alarm feed discards while safeguarding livestock.
- **Hybrid AI:** Multimodal Gemini 1.5 Flash + Deterministic Statutory Rule Engines (BIS IS:2052).

**Novelty:** First solution merging on-device computer vision with statutory Indian dairy standards.

## ► 1. Technologies to be Used (Architecture Stack & Hardware)

### ● Client & Edge Vision

- **TensorFlow.js (tfjs)**: Quantized MobileNetV1 (~1.8MB weights, WebGL backend) for 30ms sanity filtering.
- **HTML5 Canvas 2D**: In-browser pixel extraction & relative CIELAB color clustering.
- **CIEDE2000 Metric**: Sub-pixel pH colorimetry calculation.

### ● Cloud AI & Reasoning

- **Google Gemini 1.5 Flash**: Multimodal vision API for nuanced texture, chop length & spoilage parsing.
- **Structured JSON Schema**: Deterministic output mapping to avoid hallucinations.
- **Statutory Rules Engine**: TypeScript parser for BIS IS:2052:2009 standards.

### ● Progressive Web App (PWA)

- **React 18 + Vite + TypeScript**: Ultra-lean 362 kB production bundle.
- **Service Worker (sw.js)**: Full runtime & static precaching for 100% offline access.
- **IndexedDB (IDB)**: Robust queued sync for offline farm scans.

### ● Audio & Hardware Footprint

- **Web Speech Synthesis API**: Zero-network-latency vernacular voice advisory.
- **Tailwind CSS & Lucide Icons**: High-contrast responsive farm UI.
- **Zero Extra Hardware**: Operates entirely on standard sub-₹8,000 Android phones.

## ► 2. Methodology & Process for Implementation (End-to-End Operational Pipeline)

### STAGE 1

#### Capture & Reference Card Normalization

- Farmer captures feed/silage sample or pH test strip via phone camera.
- **Reference Card Auto-Calibration**: Normalizes white balance & exposure to counter extreme barn shadows.
- ROI segmentation crops strip pads and feed surface clusters.

### STAGE 2

#### On-Device Sanity Check (MobileNetV1)

- Local TFJS model executes in **<35 ms** on phone GPU/CPU.
- Screens ImageNet agricultural classes (hay, straw, corn, ear/spike).
- **Gatekeeper Guard**: Confirms image is genuine feed before consuming mobile data; blocks non-feed uploads.

### STAGE 3

#### Dual-Triage Analysis Engine (Online/Offline)

- **Online Flow (4G/WiFi)**: Gemini 1.5 Flash assesses physical characteristics, computing Flieg Score & nutritive density.
- **Offline Flow (0 kbps)**: Relative color-clustering detects mold anomalies; CIEDE2000 computes pH; results queued in IndexedDB.

### STAGE 4

#### Grading, Scorecard & Background Sync

- Scores sample against **BIS IS:2052**: Tier A (Safe), Tier B (Caution), Tier C (Reject).
- Synthesizes localized audio advisory in farmer's mother tongue.
- **Auto-Sync Engine**: Seamlessly syncs queued scans with cloud upon network reconnection.

## ► 1. Analysis of Feasibility (Technical, Operational & Economic Viability)

### TECHNICAL FEASIBILITY

#### Tested & Lightweight Architecture

- **362 kB Client Bundle:** Instant loading even on 2G/3G mobile networks.
- **Sub-100ms Inference:** Quantized MobileNetV1 & CIEDE2000 math execute smoothly on low-end MediaTek/Snapdragon chipsets.
- **PWA Deployment:** Zero app store friction; runs on any web browser.

### OPERATIONAL FEASIBILITY

#### Farmer-Centric & Zero Barrier

- **Zero Specialized Hardware:** Requires only the farmer's existing smartphone camera & inexpensive universal pH strips.
- **Simple 2-Step UX:** Click photo → Listen to voice verdict.
- **Cooperative Ready:** Seamlessly fits dairy society milk collection centers.

### ECONOMIC VIABILITY

#### Massive Cost Advantage & High ROI

- **<₹0.50 Per Test** vs ₹1,500–₹2,500 for commercial lab assays (>99% cost reduction).
- **Rapid ROI:** 15–20% milk yield improvement translates to ₹2,500+ extra monthly income per cow.
- **Low Cloud Overhead:** Edge sanity filter avoids unnecessary API costs.

## ► 2 & 3. Potential Challenges, Risks & Mitigation Strategies

### POINTER 2

#### Potential Challenges and Risks

- **Lighting & Shadow Variability:** Cattle sheds have erratic lighting (harsh sunlight, dim incandescents, heavy shadows) that can skew strip colorimetric readings.
- **Zero Rural Connectivity:** Remote dairy farms and silo pits often experience complete mobile network blackouts, risking app failure.
- **Risk of False Mold Reassurance / False Alarm:** Pure absolute color thresholds can falsely flag naturally pale straw/bhusa as mold or miss subtle early-stage fungal mycotoxins.
- **Sensor Inconsistency:** Cheap uncalibrated smartphone cameras have differing white-balance algorithms and color saturations.

### POINTER 3

#### Strategies for Overcoming These Challenges

- **Reference Card White-Balance Correction:** Automated background normalization anchors RGB channels using known neutral color patches.
- **Robust Offline-First Fallback:** Service Worker runtime precaching and IndexedDB persistent storage ensure 100% functionality without internet.
- **Relative Clustering & Tier B Safety Bounds:** Mold detection measures divergence relative to the sample's dominant color cluster; offline scans are strictly capped at Tier B (Caution) with clear visual disclaimers.
- **CIEDE2000 Perceptual Metric:** Standardized Lab color delta calculation ensures consistent color matching regardless of phone brand.



# IMPACT AND BENEFITS



## ► 1. Potential Impact on the Target Audience

| 80M+ SMALLHOLDER FARMERS                                                                                                                                                                                                                                                                                                                                    | DAIRY COOPERATIVES & SOCIETIES                                                                                                                                                                                                                                                                                                            | DAIRY REGULATORS & VETS                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Livelihood &amp; Herd Protection</b> <ul style="list-style-type: none"><li>• Democratizes laboratory-grade feed analysis into a simple 30-sec mobile scan.</li><li>• Prevents silent production losses from poorly fermented silage and aflatoxicosis.</li><li>• Voice guidance in local dialects enables fully autonomous use by all farmers.</li></ul> | <b>Village Collection Quality Gates</b> <ul style="list-style-type: none"><li>• Provides AMUL, Nandini &amp; Mother Dairy collection centers an objective quality audit.</li><li>• Incentivizes farmers with quality-based silage premiums.</li><li>• Prevents contaminated milk pooling at the Bulk Milk Chilling (BMC) level.</li></ul> | <b>Statutory Compliance &amp; Health</b> <ul style="list-style-type: none"><li>• Automated validation against BIS IS:2052:2009 feed compounding standards.</li><li>• Eliminates veterinary emergencies from acute acidosis &amp; digestive disorders.</li><li>• Cloud analytics map regional feed quality trends for DAHD interventions.</li></ul> |

## ► 2. Triple-Bottom-Line Benefits of the Solution

| ECONOMIC BENEFITS                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SOCIAL BENEFITS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ENVIRONMENTAL BENEFITS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Income Growth &amp; Asset Preservation</b> <ul style="list-style-type: none"><li>• <b>+15% to 20% Higher Milk Yield:</b> Scientifically optimized forage feeding boosts peak lactation and solids-not-fat (SNF) percentages.</li><li>• <b>₹15,000–₹25,000 Saved Per Cow:</b> Avoids emergency vet treatments, liver damage, and premature culling caused by mycotoxins.</li><li>• <b>85% Direct Testing Savings:</b> Eliminates expensive courier and lab fees for feed tests.</li></ul> | <b>Public Health &amp; Women Empowerment</b> <ul style="list-style-type: none"><li>• <b>Aflatoxin M1 Reduction in Milk:</b> Directly protects millions of children and consumers from carcinogenic toxins in dairy products.</li><li>• <b>Women Dairy Farmer Empowerment:</b> 70% of dairy management is handled by women; intuitive audio prompts bridge technological barriers.</li><li>• <b>Equitable Dairy Economy:</b> Levels the playing field between small rural farms and industrial dairy conglomerates.</li></ul> | <b>Methane Reduction &amp; Zero Waste</b> <ul style="list-style-type: none"><li>• <b>30% Reduction in Silage Spoilage:</b> Timely spoilage alerts prevent massive post-harvest biomass waste in open pit silos.</li><li>• <b>8%–12% Enteric Methane Cut:</b> Balanced digestible protein and optimal silage pH enhance rumen fermentation efficiency, directly lowering methane emissions per liter of milk.</li><li>• <b>Zero Chemical Waste:</b> Eliminates hazardous reagents used in traditional Kjeldahl and Soxhlet lab assays.</li></ul> |

## STATUTORY BENCHMARKS

### Indian Regulatory Standards

- **[BIS 2009]** Bureau of Indian Standards IS:2052:2009: *Compounded Feeds for Cattle – Specification (Fourth Revision)*. Governs mandatory thresholds for Moisture ( $\leq 11\%$ ), Crude Protein ( $\geq 20\%$  Type I,  $\geq 18\%$  Type II), Crude Fat ( $\geq 2.5\%$ ), and Acid Insoluble Ash ( $\leq 3.5\%$ ).
- **[FSSAI 2011]** Food Safety and Standards Authority of India (FSSAI): *Food Safety and Standards (Contaminants, Toxins & Residues) Regulations, 2011*. Establishes the  $0.5 \mu\text{g/kg}$  statutory limit for Aflatoxin M1 in dairy milk.
- **[ICAR-NDRI 2013]** National Dairy Research Institute, Karnal: *Nutrient Requirements of Cattle & Buffaloes*. Rumen fermentation standards and dry matter intake benchmarks.
- **[NDDB RBP]** National Dairy Development Board: *Ration Balancing Programme Field Manual* for optimized silage feeding.

## SCIENTIFIC LITERATURE

### Peer-Reviewed Scientific Foundations

- **[Flieg 1938]** Flieg, O. (1938): "A Key for the Evaluation of Silage", Futterbau und Gärfutterbereitung, 1:112–128. Foundation for the algorithmic calculation of silage fermentation quality index based on pH and lactic-to-butyric acid ratios.
- **[CIE 2001]** CIE Technical Report 142-2001: "Improvement to Industrial Colour-Difference Evaluation" (CIEDE2000 standard). Rigorous mathematical model for perceptual color distance ( $\Delta E^*00$ ) used in phone camera pH strip reading.
- **[Howard 2017]** Howard, A. G., et al. (2017): "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications", arXiv:1704.04861. Edge vision model enabling 30ms on-device feed classification.
- **[Whitlow 2005]** Whitlow, L. W., & Hagler, W. M. (2005): "Mycotoxins in Dairy Cattle", Proc. Southwest Nutrition Conference. Impact of feed molds on milk yield and cow reproduction.

## OPEN SOURCE & CODEBASE

### Project Repository & Technical Assets

- **[GITHUB REPO]** Official PashuPoshan AI Repository: [github.com/kmrpulkit006-web/PashuPoshan](https://github.com/kmrpulkit006-web/PashuPoshan)  
Fully open-source, production-ready PWA codebase with CI/CD passing on Ubuntu runners.
- **[W3C PWA]** W3C Service Worker & IndexedDB APIs: Off-grid local state synchronization specifications for zero-connectivity operation.
- **[GOOGLE AI]** Google DeepMind Gemini 1.5 Flash API: Multimodal structured extraction and low-latency agricultural vision reasoning.
- **[TEST SUITE]** Automated Verification Suite: 35+ end-to-end tests covering CIEDE2000 math, on-device vision gating, and offline triage sync.